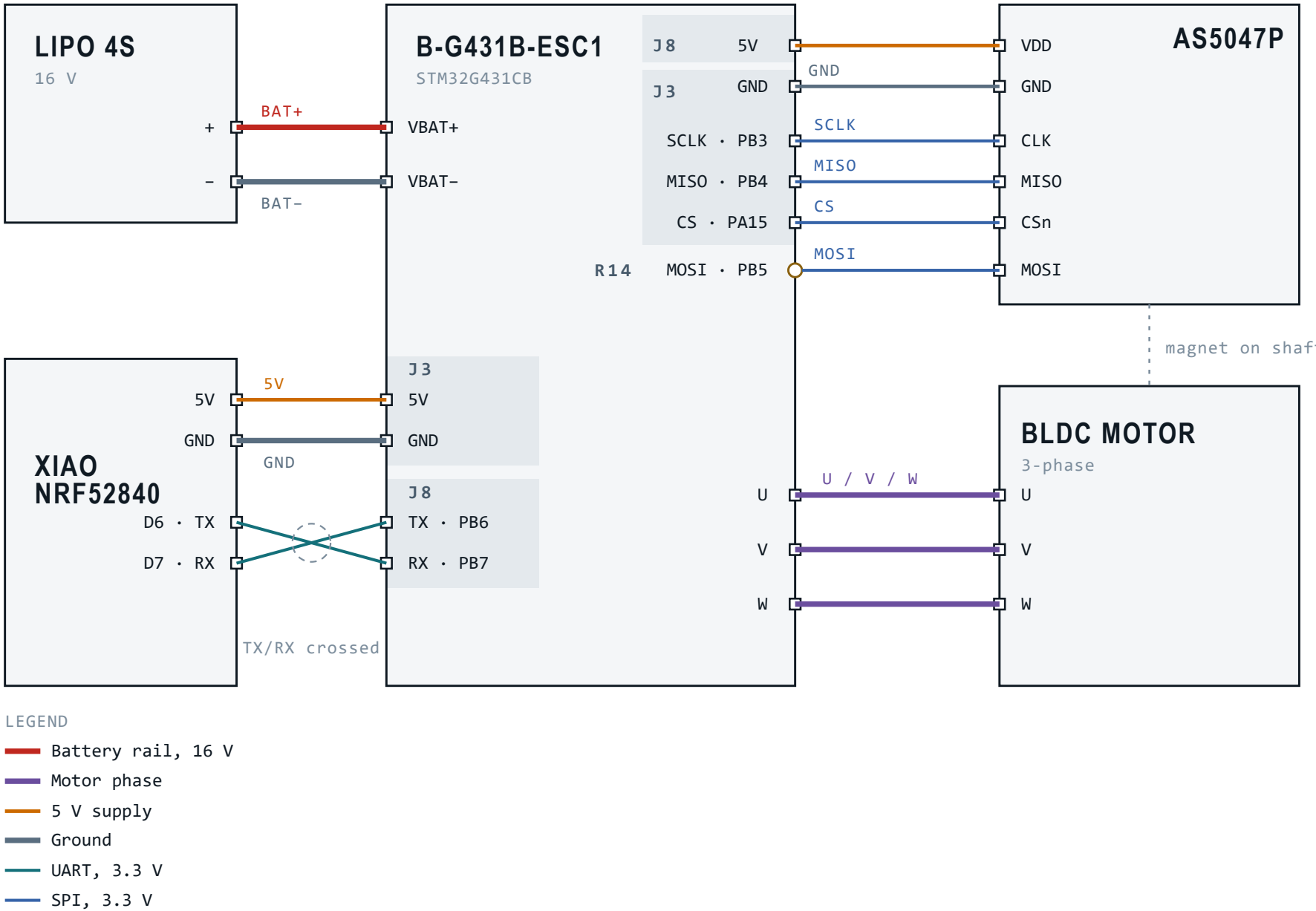

LIPO 4S
16 V
+
−
BAT+
BAT−
B-G431B-ESC1
STM32G431CB
VBAT+
VBAT−
J8
5V
J3
GND
SCLK · PB3
MISO · PB4
CS · PA15
R14
MOSI · PB5
AS5047P
VDD
GND
CLK
MISO
CSn
MOSI
SCLK
MISO
CS
MOSI
magnet on shaft
XIAO
NRF52840
5V
GND
D6 · TX
D7 · RX
5V
GND
J3
5V
GND
J8
TX · PB6
RX · PB7
TX/RX crossed
BLDC MOTOR
3-phase
U
V
W
U / V / W
LEGEND
Battery rail, 16 V
Motor phase
5 V supply
Ground
UART, 3.3 V
SPI, 3.3 V